



Program: BSc in Software Engineering

Exam Type: JU-Mock Exit Exam: Round-1

Exam Item: Multiple Choice

Number of Questions: 100

Time Allowed: 3:00hrs

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Jimma University

1. Which of the following *best* describes a **PROGRAMMING LANGUAGE**?

- A. Designing and writing instructions for computers
- B. To translate the entire source code before execution
- C. A formal language used to write programs
- D. A set of instructions written to perform tasks

ANSWER: C

2. Consider the C++ code snippet below. What will be the value of **y** after executing the following code?

```
#include<iostream>
using namespace std;
int main() {
    int y, a=4, b=8, c=10, d=7;
    y= a-- + b++ - ++c + d++;
    return 0;
}
```

- A. 8
- B. 7
- C. 6
- D. 0

ANSWER: A

3. Among the following, which loop is commonly used when the number of *iterations* is known?

- A. while loop
- B. do-while loop
- C. for loop
- D. nested loop

ANSWER: C

4. Which of the following is NOT a user-defined data type?

- A. struct
- B. enum
- C. int
- D. class

ANSWER: C

5. Consider the C++ code snippet below. What will be the output after executing the following code?

```
#include <iostream>
using namespace std;
int main() {
    int a, b, c;
    a = 8, b = 6;
    c = (a << 2) | (b >> 1);
    cout << c;
    return 0;
}
```

- A. 32
- B. 35
- C. 34
- D. 36

ANSWER: B

6. Consider the C++ code snippet below: The function `update()` modifies the value of **num** in the `main()` function. Which parameter passing method is used?

```
#include <iostream>
using namespace std;
void update (int &x) {
    x = x + 10;
}
int main() {
    int num = 5;
    update(num);
    cout << num;
    return 0;
}
```

- A. Pass by value
- B. Pass by reference
- C. Pass by constant only
- D. Pass by copy

ANSWER: B

7. Review the following conditional evaluation statement:

```
result = (x > 10) ? (y < 5 ? 1 : 2) : 3;
```

If $x = 15$ and $y = 8$, what value is assigned to the variable `result`?

- A. 2
- B. 1
- C. 3
- D. 15

ANSWER: A

8. Which explicit file open mode flag should be passed to an **`std::ofstream`** constructor to ensure that existing file content is preserved and new records are appended to the end of the stream?

- A. `std::ios::in`
- B. `std::ios::trunc`

C. `std::ios::out`

D. `std::ios::app`

ANSWER: D

9. Imagine a software developer configures a loop statement below:

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 0; i < 10; i++) {
        if (i == 5)
            continue;
        cout << i;
    }
    return 0;
}
```

Which of the following describes the accurate **output trace** behavior of this processing block?

- A. The program prints numbers from 0 to 4, then completely breaks out of the loop construct.
- B. The loop skips printing the digit 5 specifically, but successfully executes processing for all other indices.
- C. The execution engine halts execution entirely, logging a critical runtime memory boundary exception.
- D. The loop loops infinitely at index value 5 because the loop control increment step is bypassed.

ANSWER: B

10. Consider the following structural array declaration sequence:

```
int matrix[2][3] = {{1, 2, 3}, {4, 5, 6}};
```

What is the explicit evaluation outcome when looking up the record position matrix [1][0]?

- A. 1
- B. 2
- C. 4
- D. 5

ANSWER: C

11. Suppose freshman students' practices writing simple C++ programs every day without being required by the instructor because they want to improve their skills. This behavior shows:

- A. External pressure
- B. Self-motivation
- C. Lack of understanding
- D. Memorization only

ANSWER: B

12. Which of the following is the name for all computers connected to a network that participate directly in network communication?

- A. Media
- B. Hosts
- C. Servers
- D. Intermediary devices

ANSWER: B

13. What are the two sublayers of the OSI model data link layer?

- A. LLC and MAC
- B. NIC and MAC
- C. Physical and LLC
- D. Transport and Physical

ANSWER: A

14. Which statement is true about the TCP/IP and OSI models?

- A. The TCP/IP transport layer and OSI Layer 4 provide similar services and functions.
- B. The TCP/IP network access layer has similar functions to the OSI network layer.
- C. The OSI Layer 7 and the TCP/IP application layer provide identical functions.
- D. The first three OSI layers describe general services that are also provided by the TCP/IP internet layer.

ANSWER: A

15. Which of the following network device repeats incoming data frames only to the computers to which a frame is addressed by using MAC address?

- A. Active hub
- B. Router
- C. Switch
- D. Firewall

ANSWER: C.

16. Which PDU format is used when bits are received from the network medium by the NIC of a host?

- A. File
- B. Frame
- C. Packet
- D. Segment

ANSWER: B

17. Which message delivery option is used to when all deices need to receive the same message simultaneously?

- A. Duplex
- B. Unicast
- C. Multicast
- D. Broadcast

ANSWER: D.

18. Why low coupling and high cohesion are desirable in software engineering?

- A. Increase hardware dependency
- B. Improve maintainability and flexibility
- C. Eliminate the need for testing
- D. Prevent software documentation

ANSWER: B.

19. Which of the following statement explains the relationship between software quality and Software Engineering principles?

- A. Software quality depends only on programming language selection
- B. Applying Software Engineering principles improves software quality attributes
- C. Software quality eliminates the need for maintenance
- D. Software quality depends only on hardware performance

ANSWER: B.

20. Why incremental models are *advantageous*?

- A. They reduce risk by delivering the full system at once.
- B. They allow partial implementation to be deployed and evaluated early.
- C. They eliminate the need for documentation.
- D. They strictly follow sequential phases.

ANSWER: B.

21. For a large-scale enterprise system with high technical risks, changing requirements, and strict quality concerns, the most suitable model would be:

- A. Waterfall model
- B. Agile model
- C. Spiral model
- D. RAD model

ANSWER: C.

22. Which of the following software process activity includes correcting faults after software deployment?

- A. Requirement analysis
- B. System design
- C. Maintenance
- D. Validation

ANSWER: C.

23. Which of the following software Engineering principle primarily helps minimize the impact of changes in one module on other modules?

- A. Redundancy
- B. High coupling
- C. Low cohesion
- D. Low coupling

ANSWER: D.

24. Which type of malware disguises itself as legitimate software to trick users into installing it, allowing attackers to gain unauthorized access to a system?

- A. Spyware
- B. Trojan Horse
- C. Ransomware
- D. Worm

ANSWER: B.

25. How do firewalls protect networks?

- A. By isolating specific applications from each other
- B. By detecting and alerting on malicious activities
- C. By encrypting all outgoing data packets
- D. By blocking unauthorized access based on predefined rules

ANSWER: D.

26. Which cryptographic technique converts data into a fixed-length value and is commonly used to verify data integrity rather than to encrypt data?

- A. Hashing
- B. Asymmetric cryptography
- C. Symmetric cryptography
- D. Public Key Infrastructure

ANSWER: A

27. Which cryptographic algorithm relies on the mathematical difficulty of factoring large prime numbers and is widely used for secure data transmission?

- A. RSA: Rivest–Shamir–Adleman
- B. DES: Data Encryption Standard
- C. ECC: Elliptic Curve Cryptography
- D. MD5: Message Digest Algorithm 5

ANSWER: A

28. A Software Engineer is designing a web application where users authenticate with a password and receive a code via email for additional security. What type of *authentication* is being implemented?

- A. Single-Factor Authentication
- B. Biometric Authentication
- C. Multi-Factor Authentication
- D. Token-Based Authentication

ANSWER: C

29. Which firewall type examines individual packets based on information such as *source IP address*, *destination IP address*, and *port number* without considering connection state?

- A. Proxy Firewall
- B. Stateful Inspection Firewall
- C. Packet Filtering Firewall
- D. Next-Generation Firewall

ANSWER: C

30. Which condition occurs when a process is holding at least one resource while simultaneously waiting to acquire additional resources held by other processes?

- A. Mutual Exclusion
- B. Hold and Wait
- C. Thrashing
- D. Circular Wait

ANSWER: B

31. An operating system divides both physical and logical memory into *fixed-size blocks* and maps them using page tables.

- A. Virtual Memory
- B. Segmentation
- C. Swapping
- D. Paging

ANSWER: D.

32. In file management, which technique temporarily stores data in memory to improve *input/output performance* by reducing the speed gap between CPU and storage devices?

- A. Buffering
- B. Encryption
- C. RAID
- D. File Permissions

ANSWER: A.

33. Assume there are 5 processes scheduled using *Round Robin Scheduling* algorithm with a time quantum of **2 ms**. All processes arrive at time 0 ms. The burst times are given below:

Process	Burst Time
P1	6 ms
P2	5 ms
P3	2ms
P4	3 ms
P5	7 ms

Which process completes first?

- A. P1
- B. P2
- C. P3
- D. P5

ANSWER: C

34. Which of the following is a key method for *managing data security* issues?

- A. Ignoring small vulnerabilities
- B. Patch management
- C. Increasing the size of data storage
- D. Using only symmetric encryption

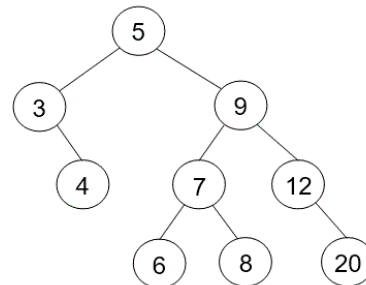
ANSWER: B

35. Which file management issue arises when multiple processes attempt to access the same file simultaneously?

- A. Fragmentation
- B. File Corruption
- C. Race Condition
- D. Deadlock

ANSWER: C

36. The binary search tree shown below was constructed by inserting a sequence of items into an empty tree. Which of the following input sequences will not generate the above binary search tree?



- A. 5, 3, 4, 9, 12, 7, 8, 6, 20
- B. 5, 9, 3, 7, 6, 8, 4, 12, 20
- C. 5, 9, 7, 8, 6, 12, 20, 3, 4
- D. 5, 9, 3, 6, 7, 8, 4, 12, 20

Answer D

37. Which of the following best defines complexity analysis?
- A. The process of determining the exact time required for an algorithm
 - B. The systematic study of the cost of computation measured in time units or operations
 - C. The study of how programming languages affect algorithm performance
 - D. The analysis of how many programmers are needed to implement an algorithm

ANSWER B

38. In **Bubble Sort**, what happens after the first complete traversal through the array?
- A. The largest element is placed in the last position
 - B. The smallest element is placed in the first position
 - C. The array is completely sorted
 - D. All adjacent elements are swapped

ANSWER A.

39. Which data structure would be most appropriate for implementing a print job scheduler where jobs are processed in the order they are received?
- A. Stack
 - B. Queue
 - C. Binary Search Tree
 - D. Heap

ANSWER B.

40. Which sorting algorithm has the same average and worst-case time bounds in **Big-Oh**?
- A. Insertion sort
 - B. Quicksort
 - C. Shellsort
 - D. Mergesort

ANSWER D.

41. Which traversal algorithm uses a stack data structure to explore a graph as deeply as possible before backtracking?
- A. Level Order Traversal
 - B. Inorder Traversal
 - C. Breadth-First Search (BFS)
 - D. Depth-First Search (DFS)

ANSWER C.

42. Which PHP super global provides data that includes values from both `$_GET`, `$_POST`, and `$_SESSION`?

- A. `$_GLOBAL`
- B. `$_INPUT`
- C. `$_REQUEST`
- D. `$_DATA`

ANSWER C.

43. When uploading a file with PHP, which attribute must be present in the form tag?

- A. `enctype="application/json"`
- B. `enctype="multipart/form-data"`
- C. `type="upload"`
- D. `method="upload"`

ANSWER B.

44. Choosing POST instead of GET for form submission provides:

- A. Faster processing
- B. URL-based data sharing
- C. Unlimited data size & improved security
- D. Ability to execute JavaScript

ANSWER C.

45. The characteristic that each request a browser sends to an HTTP server is treated independently of any previous request is known as:

- A. Encrypted
- B. Stateful
- C. Layered
- D. Stateless

ANSWER D.

46. Which HTML element is used to create a multi-line, scrollable text entry box in a form?

- A. `<input type="text">`
- B. `<input type="multiline">`
- C. `<textarea>`
- D. `<input type="comment">`

ANSWER C.

47. Which attribute, when added to a form input field, ensures that the user must enter a value before the form can be submitted?

- A. `validate`
- B. `check`
- C. `mandatory`
- D. `required`

ANSWER D.

48. Which CSS unit is a relative length unit that is based on the font size of the element's parent element?

- A. `vw` (viewport width)
- B. `rem` (root em)
- C. `em`
- D. `px` (pixels)

ANSWER C.

49. Given the following CSS, what is the final color of `<p>` inside `<div>`?

```
p { color: red; }  
div p { color: blue; }  
#main p { color: green; }
```

If the `<div>` has `id="main"`, the text color will be:

- A. red
- B. blue
- C. green
- D. undefined

ANSWER C.

50. Which of the following rules will override all others, assuming equal specificity?

- A. Inline style
- B. External stylesheet
- C. Embedded stylesheet
- D. Browser default

ANSWER A.

51. Which one of the following is evaluating the degree of success of an agent?

- A. Knowledge
- B. Perception
- C. Performance measure
- D. Action

ANSWER C

52. Which search strategy is guaranteed to find the shortest path if step costs are identical?

- A. Breadth-First Search
- B. Depth-First Search
- C. Greedy Best-First Search
- D. Iterative Deepening Search

ANSWER: A

53. To solve the **8-Puzzle Problem**, you decide to use a heuristic search strategy. Which algorithm would combine the actual cost and estimated cost to the goal state?

- A. Breadth-First Search
- B. Uniform-Cost Search
- C. A* Search
- D. Greedy Best-First Search

ANSWER: C

54. A smart home system follows the rule: *"If motion is detected, then the lights turn on."* Which type of logic can represent this rule as $P \rightarrow Q$?

- A. Fuzzy Logic
- B. Predicate Logic
- C. Propositional Logic
- D. Semantic Networks

ANSWER: C.

55. Why is lifelong learning important for **AI** professionals?

- A. To replace all existing knowledge periodically
- B. To adapt to evolving new technologies

- C. To focus only on programming languages
- D. To eliminate dependency on automated tools

ANSWER: B.

56. Which of the following is NOT a type of unsupervised learning task?

- A. Classification
- B. Clustering
- C. Dimensionality reduction
- D. Anomaly detection

ANSWER: A.

57. Imagine have collected data on your daily exercise routines and calorie intake. Your goal is to build a model that ***predicts*** how many calories you should burn based on your planned exercise. What type of machine learning should you use?

- A. Unsupervised Learning,
- B. Supervised Learning
- C. Reinforcement Learning
- D. Deep Learning

ANSWER: B.

58. Which of the following is a NON-LINEAR classification model?

- A. Linear Regression
- B. Logistic Regression
- C. Support Vector Machines
- D. K-Nearest Neighbors (KNN)

ANSWER: D.

59. You have historical data on student performance, including demographics, study habits, and past grades. Your goal is to build a model that identifies students at risk of dropping out. What classification algorithm is best suited for this task?

- A. Random Forest
- B. K-Means Clustering
- C. Naive Bayes
- D. Principal Component Analysis (PCA)

ANSWER: A.

60. Which evaluation metric is NOT used for regression models?

- A. Mean squared error (MSE).
- B. Root mean squared error (RMSE).
- C. Mean absolute error (MAE).
- D. Precision and recall.

ANSWER: D.

61. What is the main purpose of the AndroidManifest.xml file?

- A. To store application images
- B. To write the application business logic
- C. To declare application components
- D. To display user interfaces

ANSWER: C.

62. Which Android component runs in the background without a user interface?

- A. Service
- B. Activity
- C. Fragment
- D. View

ANSWER: A.

63. A student creates a new Android project in Android Studio and wants to specify the minimum Android version that can run the application. Which file should the student configure?

- A. strings.xml
- B. activity_main.xml
- C. drawable folder
- D. AndroidManifest.xml

ANSWER: D

64. A user enters their username into an *EditText field*. Which method should be used to retrieve the entered value?

- A. `getValue()`
- B. `getData()`
- C. `getText().toString()`
- D. `findText()`

ANSWER: C

65. Which Android component is used to request an action from another component, such as launching an activity or starting a service?

- A. Service
- B. Broadcast Receiver
- C. Activity
- D. Intent

ANSWER: D

66. Which project management model is best suited for a project with constantly evolving requirements?

- A. Waterfall model: rigid and sequential
- B. Agile development: flexible and adaptable
- C. Spiral model: risk-driven with iteration
- D. Lean software development: focuses on minimizing waste

ANSWER: B

67. When estimating the cost of a software project, which of the following is the LEAST reliable technique?

- A. Expert judgment from experienced project managers.
- B. Analogous estimating based on similar past projects.
- C. Parametric estimating using historical data and formulas.
- D. Guessing without any data or prior experience.

ANSWER: D

68. What is the PRIMARY purpose of a Work Breakdown Structure (WBS) in software development?

- A. To define the technical architecture of the software.
- B. To decompose the project into manageable tasks and deliverables.
- C. To document user interface (UI) mockups and design elements.
- D. To create a list of potential risks associated with the project.

ANSWER: B

69. Which document shows *project activities, durations, and dependencies*?

- A. Schedule plan
- B. User guide
- C. Test report
- D. Risk log

ANSWER: A

70. A software development project for a new e-commerce platform is underway using a Waterfall methodology. The project manager notices the team is falling behind schedule due to unforeseen technical challenges integrating with a third-party payment gateway. What is the MOST appropriate action for the project manager to take in this situation?

- A. Ignore the schedule delay and focus on completing the integration.
- B. Increase the project budget to cover potential overtime costs.
- C. Update the project schedule to reflect the realistic timeline for integration.
- D. Delegate the issue to the development team to resolve without informing stakeholders.

ANSWER: C

71. What is the PRIMARY benefit of conducting code reviews in software development?

- A. Reduce development time of the project
- B. Identify potential bugs and code smells
- C. Increase code complexity
- D. Automate unit testing

ANSWER: B

72. During a software review, you are analyzing a software application for potential maintenance challenges. The application has a complex codebase with numerous code smells and lacks proper documentation. What is the most likely consequence?

- A. Low maintenance effort and cost
- B. Increased maintenance effort and cost
- C. No impact on future maintenance
- D. Potential for security vulnerabilities only

ANSWER: B

73. Which maintenance approach should be applied to modernize **outdated software** technology?

- A. Reengineering
- B. Corrective maintenance
- C. Preventive testing
- D. Data backup

ANSWER: A

74. Which software maintenance plan should emphasize periodic code reviews and refactoring?

- A. Preventive Maintenance Plan
- B. Corrective Maintenance Plan
- C. Perfective maintenance plan
- D. Deployment Plan

ANSWER: A

75. Which of the following factors is NOT typically considered when selecting a version control system for software maintenance?

- A. Ease of use and collaboration features
- B. Scalability to handle large codebases
- C. Integration with bug tracking systems
- D. The programming language used for the software

ANSWER: D

76. Consider the JAVA code snippet below. What will be the output after executing the following code?

```
Public class Test1 {  
    Public static void main(String args []){  
        for (int i = 1; i <= 5; i++) {  
            if (i == 3)  
                break;  
            System.out.print(i + " ");  
        }  
    }  
}
```

- A. 1 2 3 4 5
- B. 1 2
- C. 1 2 3
- D. Infinite loop

ANSWER: B

77. Which keyword is commonly used to refer to the current object?

- A. this
- B. super
- C. new
- D. object

ANSWER: A

78. Which Swing container is commonly used as the main window of a desktop application?

- A. JRootPane
- B. JPanel
- C. JFrame
- D. JMenuBar

ANSWER: C

79. A class Car extends a class Vehicle to reuse its properties and methods. Which concept is demonstrated?

- A. Encapsulation
- B. Inheritance
- C. Polymorphism
- D. Abstraction

ANSWER: B

80. Which of the following classes is a checked exception in Java, meaning the compiler forces you to handle it or declare it in a throws clause?

- A. NullPointerException
- B. ArithmeticException
- C. IllegalArgumentException
- D. ClassNotFoundException

ANSWER: D

81. Which database model organizes data in a *parent-child structure*?

- A. Relational model
- B. Network model
- C. Hierarchical model
- D. Object-oriented model

ANSWER: C

82. Which command removes all records from a table but keeps its structure?

- A. DROP
- B. DELETE
- C. TRUNCATE
- D. REMOVE

ANSWER: C

83. What will be the result of executing the SQL statement below, assuming that *id* is defined as a PRIMARY KEY?

INSERT INTO Student VALUES (null, Abdi);

- A. It will insert successfully
- B. It will update existing record
- C. It will delete the table
- D. It will cause an error

ANSWER: D

84. Which database design activity focuses on converting real-world objects into database tables?

- A. Normalization
- B. ER modeling
- C. Query optimization
- D. Data mining

ANSWER: B

85. Which type of key is used to enforce relationships between two tables?

- A. Candidate Key
- B. Foreign Key
- C. Super Key
- D. Primary Key

ANSWER: B

86. Sensitive customer data is exposed due to hacking. What is the most critical initial action?

- A. Launch marketing campaigns
- B. Ignore until investigation completes
- C. Replace the entire database immediately
- D. Identify how the attacker gained access

ANSWER: D

87. Which of the following types of testing focuses on ensuring that new changes in code do not adversely affect existing functionality?

- A. Unit Testing
- B. Regression Testing
- C. Performance Testing
- D. Smoke Testing

ANSWER: B

88. Which of the following tests is performed to ensure that the system meets user requirements and business objectives?

- A. Unit Testing
- B. System Testing
- C. Acceptance Testing
- D. Stress

ANSWER: C

89. A software team is developing a test approach for a new banking system. Which document best combines scope, resources, risks, and testing approach?

- A. Test Case
- B. Test Plan
- C. Defect Report
- D. Use Case

ANSWER: B

90. A test strategy defines how both manual and automated testing will be used. What is the main benefit of this decision?

- A. Reduces requirement gathering
- B. Eliminates defects completely
- C. Removes need for test cases
- D. Improves test coverage and efficiency

ANSWER: D

91. A software project team identifies potential system failure risks during development. What should be the first step in managing these risks?

- A. Identify and analyze the risks
- B. Ignore minor risks
- C. Deploy the system immediately
- D. Increase project budget

ANSWER: A

92. . What makes requirement engineering an important stage in the software development process?

- A. It is where coding is done
- B. It improves the graphical interface of the project
- C. It deals with software and hardware components
- D. It forms the foundation for design and development.

ANSWER: D

93. Which requirement elicitation technique helps identify requirements by watching users in their real working environment?

- A. Survey
- B. Interface design
- C. Observation
- D. Testing

ANSWER: C

94. A stakeholder keeps changing their requirements after each discussion. Which elicitation technique can help solidify their ideas?

- A. Observation
- B. Prototyping
- C. Questionnaire
- D. Focus group

ANSWER: B

95. During requirement change analysis, a team evaluates *cost, schedule, and risk*. What is this activity called?

- A. Requirement coding
- B. System deployment
- C. Interface design
- D. Impact analysis

ANSWER: D

96. Which of the following is NOT a role of requirement engineering?

- A. Requirement gathering
- B. Requirement analysis
- C. Code compilation
- D. Requirement validation

ANSWER: C

97. Why is “***Don’t reinvent the wheel***” important in software design?

- A. It increases development cost
- B. It promotes reuse of existing solutions
- C. It makes software slower
- D. It removes testing needs

ANSWER: B

98. A software company is organizing its codebase into modules, packages, and libraries to make development easier. Which 4+1 architectural view is being used?

- A. Logical View
- B. Physical View
- C. Development View
- D. Process View

ANSWER: C

99. Which design strategy would best support scalability in a *large e-government system*?

- A. Monolithic architecture
- B. Single-server deployment
- C. Microservices architecture
- D. Manual processing system

ANSWER: C

100. A system is expected to serve millions of users simultaneously. Which design decision is most appropriate?

- A. Single-threaded processing
- B. Distributed system architecture
- C. Local file-based storage
- D. Manual scaling

ANSWER: B